Financial News Sentiment vs. NIFTY50 Movement

Project Report

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1 Project Overview

This project investigates whether the sentiment expressed in financial news headlines can help explain or predict the direction of the NIFTY50 index. The analysis combines financial news data with historical NIFTY50 market data. The workflow includes data cleaning, date alignment, news aggregation, rule-based sentiment analysis, feature engineering, exploratory analysis, machine learning, and a simple backtest.

The main objective was to determine whether news sentiment and recent market information provide useful signals for predicting the next-day direction of the NIFTY50.

Key Results

Usable observations	125
Training / Test split	100 / 25
Best test accuracy	44.0%
Best CV accuracy	51.25%
Buy-and-Hold return	-0.64%
Model strategy return	-0.77%

2 Changes and Contributions

I implemented and extended the project independently in a Jupyter Notebook. The main changes and contributions were:

- Cleaned and standardized the financial news and NIFTY50 datasets.
- Aligned news and market observations using trading dates.
- Grouped multiple news headlines occurring on the same date.
- Implemented rule-based sentiment scoring for financial news.
- Created sentiment-derived features including rolling averages and short-term sentiment variability.
- Created market features including lagged movement, volume change, and high-low spread.
- Constructed a next-day NIFTY50 direction target.
- Used a chronological train/test split to avoid future-data leakage.
- Implemented Logistic Regression, Random Forest, and XGBoost models.
- Performed Random Forest hyperparameter tuning using TimeSeriesSplit cross-validation.
- Evaluated predictions using accuracy, classification reports, and a confusion matrix.
- Analyzed Random Forest feature importance.
- Compared a model-based trading strategy against buy-and-hold.

3 Data and Feature Engineering

After preprocessing and feature construction, 125 observations were available for machine learning. Eight features were used:

`sentiment_score`	sentiment_ma3	sentiment_ma7	sentiment_std3
`movement_lag1`	movement_lag2	volume_change	high_low_spread

The target variable represents whether the NIFTY50 moves upward on the following trading day. The data was split chronologically into 100 training observations and 25 final test observations.

Time-aware evaluation: The first 100 observations were used for model training and validation, while the final 25 observations were kept completely unseen for the final evaluation. This prevents future observations from leaking into the training process.

4 Machine Learning Analysis

Three initial models were evaluated: Logistic Regression, Random Forest, and XGBoost. A majority-class classifier was also used as a baseline.

Table 1. Model performance on the held-out test period.

Model	Accuracy
Majority Baseline	44.0%
Logistic Regression	44.0%
Random Forest	44.0%
XGBoost	40.0%
Tuned Random Forest	**44.0%**

The initial results show that Logistic Regression and Random Forest did not improve upon the majority baseline, while XGBoost performed slightly worse. Random Forest hyperparameter tuning selected $n\_estimators = 100$, $max\_depth = 3$, and $min\_samples\_leaf = 20$.

The best time-series cross-validation accuracy was 51.25%, but the accuracy on the untouched test period remained 44.0%. Therefore, hyperparameter tuning did not produce an improvement in out-of-sample performance.

5 Feature Importance

The tuned Random Forest indicated that previous-day market movement (`movement_lag1`) was the most influential feature, followed by the seven-day sentiment moving average (`sentiment_ma7`). High-low spread and short-term sentiment variability were also relatively important.

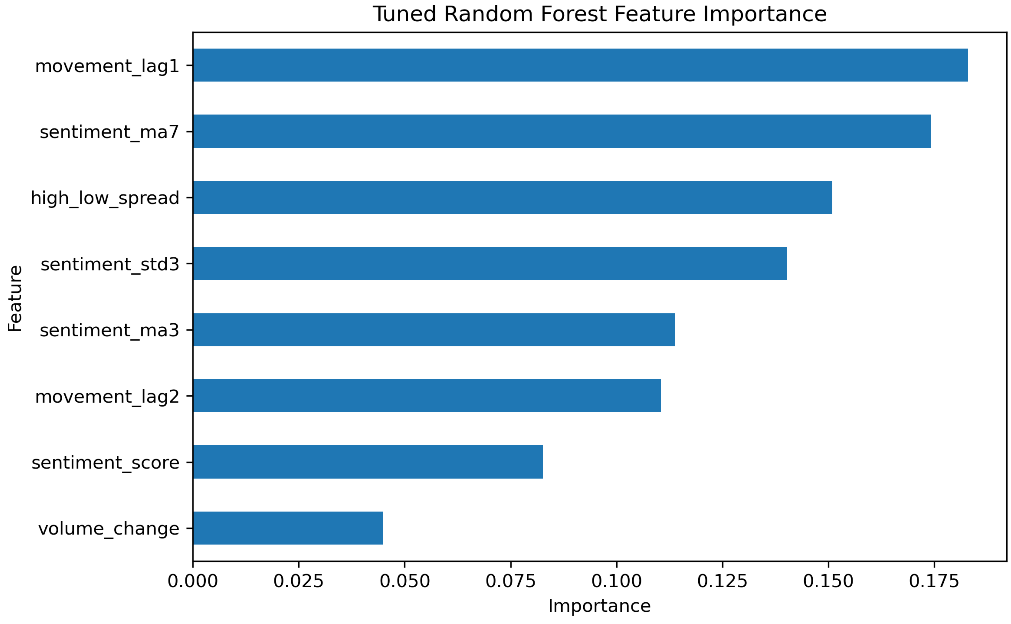

Figure 1. Feature importance from the tuned Random Forest.

These results suggest that recent market behaviour and aggregated news sentiment were more influential than raw sentiment alone within the trained Random Forest. Feature importance should be interpreted as model importance rather than evidence of causality.

6 Prediction Errors

The tuned Random Forest produced the following confusion matrix:

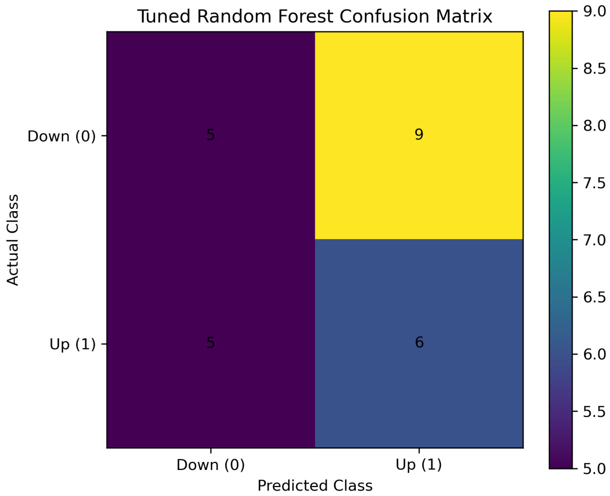

Figure 2. Confusion matrix for the tuned Random Forest.

The model correctly classified 5 downward movements and 6 upward movements, resulting in 11 correct predictions out of 25 test observations. The small test set means that one additional correct prediction changes the measured accuracy by approximately four percentage points.

7 Backtest and Discussion

A simple long-only strategy was constructed using the tuned Random Forest predictions. When the model predicted an upward movement, the strategy held the NIFTY50 for the following day; otherwise, it remained out of the market.

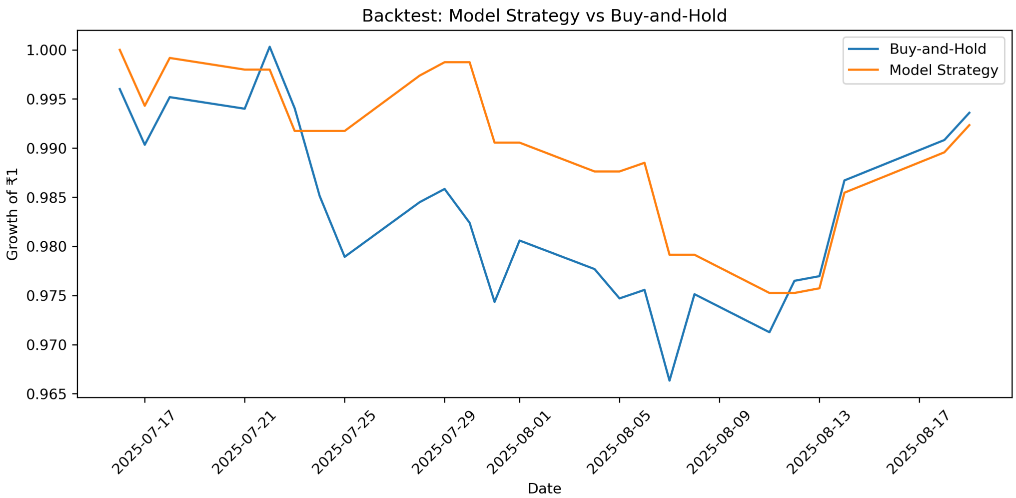

Figure 3. Model strategy compared with buy-and-hold.

Over the evaluation period, the buy-and-hold strategy returned -0.64%, while the model-based strategy returned -0.77%. Consequently, the model strategy underperformed buy-and-hold by approximately 0.13 percentage points.

8 Conclusion and Limitations

The analysis did not find evidence that the tested sentiment and market features could reliably predict next-day NIFTY50 direction during the held-out period. The best-performing models achieved 44.0% accuracy, equal to the majority baseline, while XGBoost achieved 40.0%.

The results should be interpreted cautiously because only 125 usable observations were available for machine learning and the final test set contained 25 observations. The short evaluation period means that each individual prediction has a relatively large effect on measured accuracy. The backtest also covers a limited period and does not include transaction costs, slippage, taxes, or other real-world trading effects.

Overall conclusion: The implemented models did not outperform the baseline on the held-out period. However, the project demonstrates a complete time-aware workflow from financial-news sentiment extraction and feature engineering through machine-learning evaluation and strategy comparison.